Homework 11

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11.2.2

$$I = T^{+}(\varepsilon)\,T(\varepsilon) = \left(I + \frac{i\varepsilon}{\hbar} G^{+}\right)\left(I - \frac{i\varepsilon}{\hbar} G\right)$$
$$= I + \frac{i\varepsilon}{\hbar}(G^{+} - G) + O(\varepsilon^{2})$$

$$\Rightarrow \frac{i\varepsilon}{\hbar}(G^{+} - G) = 0$$

$$\Rightarrow G^{+} = G.$$

11.4.1

$$\psi(t) = U(t)\,\psi(0)$$
$$U(t) = e^{-i\frac{H}{\hbar}t} = \sum_{n=0}^{\infty} \frac{1}{n!}\left(-i\frac{H}{\hbar}t\right)^{n}$$

Since $[\Pi, H] = 0$, $[\Pi, U(t)] = 0$.

Therefore, $\Pi\psi(t) = \Pi U(t)\psi(0) = U(t)\,\Pi\psi(0)$.

If $\Pi\psi(0) = \lambda\psi(0)$, $\lambda = \pm 1$, we have

$$\Pi\psi(t) = U(t)[\lambda\psi(0)] = \lambda\, U(t)\psi(0) = \lambda\psi(t)$$

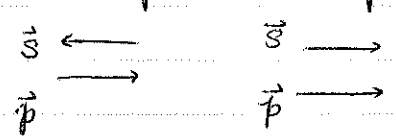
$\Rightarrow$ A system that starts out in a state of even / odd parity maintains its parity.

11.4.3

$\Pi\vec{S} = \vec{S}$, $\Pi\vec{p} = -\vec{p}$

$\Rightarrow \Pi(\vec{S}\cdot\vec{p}) = -\vec{S}\cdot\vec{p}$

Therefore, we should have $-\vec{S}\cdot\vec{p}$ and $\vec{S}\cdot\vec{p}$ with equal rate if



parity is a good symmetry. If $\vec{p}\,/\!/\,\vec{S}$, $\vec{S}\cdot\vec{p}$ is positive only, parity must be violated.

12.2.1.

$$U[R] = I - \frac{i\epsilon_z L_z}{\hbar} \quad \text{when } \epsilon_z \text{ is small.}$$

$$\langle x,y|U[R]|\psi\rangle = \iint dx'dy' \langle x,y|U[R]|x',y'\rangle\langle x',y'|\psi\rangle$$

$$= \iint dx'dy' \langle x,y|x'-y'\epsilon_z, x'\epsilon_z+y'\rangle\langle x',y'|\psi\rangle \quad (*)$$

Change of variables:

$$\begin{cases} r = x'-y'\epsilon_z \\ s = x'\epsilon_z+y' \end{cases} \Rightarrow \begin{cases} x' = \frac{1}{1+\epsilon_z^2}(r+s\epsilon_z) \approx r+s\epsilon_z \\ y' = \frac{1}{1+\epsilon_z^2}(s-r\epsilon_z) \approx s-r\epsilon_z \end{cases}$$

Jacobian: $\left|\frac{\partial(x',y')}{\partial(r,s)}\right| = 1+\epsilon_z^2 \approx 1$

Therefore,

$$(*) = \iint drds \langle x,y|r,s\rangle\langle r+s\epsilon_z, s-r\epsilon_z|\psi\rangle$$

$$= \iint drds\, \delta(x-r)\delta(y-s)\langle r+s\epsilon_z, s-r\epsilon_z|\psi\rangle$$

$$= \langle x+y\epsilon_z, y-x\epsilon_z|\psi\rangle$$

$$= \psi(x+y\epsilon_z, y-x\epsilon_z).$$

12.2.2.

First check the consistency.

$$[x, L_z] = [x, xp_y - yp_x] = -y[x,p_x] = -i\hbar y$$

$$[y, L_z] = [y, xp_y - yp_x] = x[y,p_y] = i\hbar x$$

$$[p_x, L_z] = [p_x, xp_y - yp_x] = [p_x, x]p_y = -i\hbar p_y$$

$$[p_y, L_z] = [p_y, xp_y - yp_x] = -[p_y, y]p_x = i\hbar p_x$$

Let's derive the form of L_z from the commutation relations.

$$[x, L_z] = i\hbar\frac{\partial L_z}{\partial p_x} = -i\hbar y \Rightarrow \frac{\partial L_z}{\partial p_x} = -y \quad ①$$

$$[y, L_z] = i\hbar\frac{\partial L_z}{\partial p_y} = i\hbar x \Rightarrow \frac{\partial L_z}{\partial p_y} = x \quad ②$$

$$[p_x, L_z] = -i\hbar\frac{\partial L_z}{\partial x} = -i\hbar p_y \Rightarrow \frac{\partial L_z}{\partial x} = p_y \quad ③$$

$$[p_y, L_z] = -i\hbar\frac{\partial L_z}{\partial y} = i\hbar p_x \Rightarrow \frac{\partial L_z}{\partial y} = -p_x \quad ④$$

Set $L_z = L_z(x, y, p_x, p_y)$

① $\Rightarrow L_z = -yp_x + f_1(x, y, p_y)$

② $\Rightarrow \frac{\partial L_z}{\partial p_y} = \frac{\partial f_1}{\partial p_y} = x \Rightarrow f_1 = xp_y + f_2(x,y)$

$\Rightarrow L_z = -yp_x + xp_y + f_2(x,y)$

③ $\Rightarrow \quad \frac{\partial L_z}{\partial x} = p_y + \frac{\partial f_2}{\partial x} = p_y \quad \Rightarrow \quad f_2$ is not a function of x

④ $\Rightarrow \quad \frac{\partial L_z}{\partial y} = -p_x + \frac{\partial f_2}{\partial y} = -p_x \Rightarrow \quad f_2$ is not a function of y

Therefore, f_2 can only be a const.

Hence, $\quad L_z = -y p_x + x p_y + \text{Const}.$

12.2.3

$$\begin{cases} x = \rho\cos\phi \\ y = \rho\sin\phi \end{cases}$$

$$\frac{\partial}{\partial\phi} = \frac{\partial x}{\partial\phi}\frac{\partial}{\partial x} + \frac{\partial y}{\partial\phi}\frac{\partial}{\partial y} = -\rho\sin\phi\frac{\partial}{\partial x} + \rho\cos\phi\frac{\partial}{\partial y}$$

$$= -y\frac{\partial}{\partial x} + x\frac{\partial}{\partial y}$$

Since $\quad L_z = x\left(-i\hbar\frac{\partial}{\partial y}\right) - y\left(-i\hbar\frac{\partial}{\partial x}\right)$

$$= -i\hbar\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right),$$

we have

$$L_z = -i\hbar\frac{\partial}{\partial\phi}.$$

Or directly,

$$\left(1 - i\frac{\delta\phi}{\hbar}L_z\right)\psi(\rho,\phi) = \psi(\rho,\phi-\delta\phi)$$

$$= \psi(\rho,\phi) + \frac{\partial\psi}{\partial\phi}(-\delta\phi) + O[(\delta\phi)^2]$$

$$\Rightarrow -i\frac{\delta\phi}{\hbar}L_z\psi(\rho,\phi) = \frac{\partial\psi}{\partial\phi}(-\delta\phi)$$

$$\Rightarrow L_z\psi(\rho,\phi) = -i\hbar\frac{\partial\psi}{\partial\phi}$$

$$\Rightarrow L_z = -i\hbar\frac{\partial}{\partial\phi}.$$

12.3.5

$$\left[-\frac{\hbar^2}{2\mu}\left(\frac{d^2}{d\rho^2} + \frac{1}{\rho}\frac{d}{d\rho} - \frac{m^2}{\rho^2}\right) + V(\rho)\right]R_{Em}(\rho) = E R_{Em}(\rho)$$

generated by angular momentum

$$V_{ang.} = \frac{\hbar^2 m^2}{2\mu\rho^2}$$

$$\vec{F} = -\nabla V_{ang.} = \frac{\hbar^2 m^2}{\mu\rho^3}\hat{\rho} \qquad \text{(repulsive force)}$$

Since $L_z = m\hbar$, we have $\quad F = \frac{L_z^2}{\mu\rho^3}.$

12.6.1

(1) $\psi_E(r,\theta,\phi) = A e^{-r/a_0}$ does not depend on θ, ϕ. Therefore its angular momentum is zero.

$$\hat{L}^2 = -\hbar^2 \left(\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \sin\theta \frac{\partial}{\partial\theta} + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\phi^2} \right)$$

$$\hat{L}^2 \psi_E = 0 \quad \Rightarrow \quad l = 0.$$

(2)

$$\hat{H}\psi_E = \left[-\frac{\hbar^2}{2\mu} \frac{1}{r^2} \frac{\partial}{\partial r}\left(r^2 \frac{\partial}{\partial r}\right) + V(r) \right] \psi_E = E\psi_E \quad \text{(no angular part)}$$

Substitute $\psi_E = A e^{-r/a_0}$, we have

$$E A e^{-r/a_0} = -\frac{\hbar^2}{2\mu} \frac{1}{r^2} \frac{\partial}{\partial r} \left(-r^2 \frac{A}{a_0} e^{-r/a_0}\right) + V(r) A e^{-r/a_0}$$

$$\Rightarrow \quad E e^{-r/a_0} = -\frac{\hbar^2}{2\mu} \frac{1}{r^2} \left(-2r \frac{1}{a_0} e^{-r/a_0} + \frac{r^2}{a_0^2} e^{-r/a_0} \right) + V(r) e^{-r/a_0}$$

$$\Rightarrow \quad E = \frac{\hbar^2}{a_0 \mu r} - \frac{\hbar^2}{2\mu a_0^2} + V(r) \qquad (*)$$

In the limit of $r \to \infty$, $\frac{\hbar^2}{a_0 \mu r} \to 0$, $V(r) \to 0$, we have

$$E = -\frac{\hbar^2}{2\mu a_0^2}.$$

(3) If r is finite, substitute $E = -\frac{\hbar^2}{2\mu a_0^2}$ into $(*)$, we have

$$\frac{\hbar^2}{a_0 \mu r} + V(r) = 0$$

$$\Rightarrow \quad V(r) = -\frac{\hbar^2}{a_0 \mu r}.$$

12.6.4

(1) $$\int f(\vec{r}\,') \delta^3(\vec{r} - \vec{r}\,') \, d\vec{r}\,' = f(\vec{r})$$

In Cartesian Coordinate, we have,

$$\int f(x', y', z') \delta(x - x') \delta(y - y') \delta(z - z') \, dx' dy' dz' = f(x, y, z) = f(\vec{r}).$$

In Spherical Coordinate, we have,

$$\int f(r', \theta', \phi') \frac{1}{r^2 \sin^2\theta} \delta(r - r') \delta(\theta - \theta') \delta(\phi - \phi') \, r'^2 \sin^2\theta' \, dr' d\theta' d\phi'$$

$$= \frac{1}{r^2 \sin^2\theta} f(r, \theta, \phi) \, r^2 \sin^2\theta$$

$$= f(r, \theta, \phi)$$

$$= f(\vec{r})$$

Therefore, $\delta^3(\vec{r} - \vec{r}\,') \equiv \delta(x - x') \delta(y - y') \delta(z - z') = \frac{1}{r^2 \sin^2\theta} \delta(r - r') \delta(\theta - \theta') \delta(\phi - \phi')$.

(2) If $r \neq 0$, we have

$$\nabla^2\left(\frac{1}{r}\right) = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\cdot\frac{1}{r}\right)$$
$$= \frac{1}{r^2}\frac{\partial}{\partial r}(-1)$$
$$= 0.$$

For $r \to 0$, consider a small sphere centered at the origin.

$$\int_V \nabla^2\left(\frac{1}{r}\right) dV = \int_V \nabla\cdot\left(\nabla\frac{1}{r}\right) dV$$
$$= \int_S \left(\nabla\frac{1}{r}\right)\cdot d\vec{s} \qquad \text{(Gauss's law)}$$
$$= \int \left(\nabla\frac{1}{r}\right)\cdot\left(r^2\hat{r}\,d\Omega\right)$$
$$= \int \left[\frac{\partial}{\partial r}\left(\frac{1}{r}\right)\right]\cdot r^2\,d\Omega$$
$$= \int (-1)\,d\Omega$$
$$= -4\pi.$$

Meanwhile,

$$\int_V -4\pi\,\delta^3(\vec{r})\,dV = -4\pi$$

Therefore,

$$\nabla^2\left(\frac{1}{r}\right) = -4\pi\,\delta^3(\vec{r}).$$